Ch 67 · Upper GI bleeding (non-variceal)

Risk scores

- Glasgow–Blatchford (pre-endoscopy; 0–1 outpatient); Rockall (post-endoscopy)

Before endoscopy

- Restrictive transfusion Hb 7 (8 if cardiac disease); IV PPI; erythromycin pre-scope
- Correct coagulopathy; stop NSAIDs; TXA not recommended (HALT-IT); protect airway

Endoscopy and after

- Within 24 h (12 h if unstable); Forrest Ia–IIa: dual therapy (adrenaline + clips/thermal)
- High-dose PPI 72 h; H. pylori eradication
- Rebleed: repeat endoscopy → embolisation → surgery

Key point: Over-transfusion increases rebleeding.

Ch 68 · Variceal haemorrhage

Initial care

- Airway protection; Hb target 7–8
- Terlipressin or octreotide 2–5 days; ceftriaxone up to 7 days

Endoscopy

- Within 12 h: band ligation (oesophageal); cyanoacrylate (gastric)
- Uncontrolled: balloon tamponade or oesophageal stent → rescue TIPS

Pre-emptive TIPS and prophylaxis

- Child C 10–13 or B > 7 with active bleeding: TIPS within 72 h ↑ survival
- NSBB (carvedilol) + repeat banding; lactulose; transplant assessment

Key point: Antibiotics in variceal bleeding reduce mortality – never omit them.

Ch 69 · Acute pancreatitis

Severity (revised Atlanta)

- Mild: no organ failure; moderately severe: transient (< 48 h) or local complications; severe: persistent

Management

- Moderate fluids with Ringer's lactate (WATERFALL: aggressive harmful)
- Early enteral feeding; analgesia; no prophylactic antibiotics
- ERCP only for cholangitis; same-admission cholecystectomy in mild biliary disease

Infected necrosis

- Gas on CT or deterioration after 1–2 weeks; antibiotics penetrating necrosis
- Step-up: delay ≥ 4 weeks, drain then necrosectomy (endoscopic/percutaneous)

Key point: Organ failure duration, not enzymes, defines severity.

Ch 70 · Acute liver failure

Definition and causes

- INR ≥ 1.5 + encephalopathy without cirrhosis; hyperacute/acute/subacute
- Paracetamol, viral hepatitis, DILI, autoimmune, Wilson's, ischaemic, Budd–Chiari, pregnancy

King's criteria (paracetamol)

- pH < 7.3 (or lactate > 3 after fluids), or INR > 6.5 + creatinine > 300 + grade 3–4 HE

ICU care

- NAC for all; refer early to transplant centre
- Intubate grade 3–4; Na 145–155, head-up, normocapnia; ammonia > 150 → CRRT
- Noradrenaline, glucose checks, infection screening; plasma exchange improves transplant-free survival

Key point: Cerebral oedema is the main cause of death in hyperacute failure.

Ch 71 · Lower GI bleeding

Assessment

- Diverticula, angiodysplasia, ischaemic colitis, malignancy, post-polypectomy
- Exclude upper source; shock index > 1 = unstable; Oakland score for discharge

Management

- Restrictive transfusion; reverse anticoagulants if life-threatening
- Unstable: CT angiography → embolisation; stable: colonoscopy within 24 h

Surgery

- Ongoing instability with failed IR/endoscopy; segmental resection after localisation

Key point: In unstable bleeding, CT angiography comes before colonoscopy.

Ch 73 · Nutrition in critical illness

Assessment

- NUTRIC or NRS-2002; indirect calorimetry; 20–25 kcal/kg built up; protein 1.2–2.0 g/kg

Route and timing

- Enteral within 24–48 h once stable; trophic first week (EDEN, PermiT, TARGET)
- PN delayed to day 3–7 if well nourished (EPaNIC); earlier if malnourished
- Gastric first; prokinetics, then post-pyloric

Refeeding

- Low PO4, K, Mg; thiamine first; start 10–15 kcal/kg/day and replace daily

Key point: Early full feeding gives no benefit – build up slowly.

Ch 74 · Spontaneous bacterial peritonitis

Diagnosis

- Ascitic neutrophils ≥ 250/mm³; culture in blood bottles
- Secondary peritonitis: polymicrobial, protein > 10 g/L, glucose < 2.8, high LDH → CT

Treatment

- Third-generation cephalosporin; albumin 1.5 g/kg day 1 + 1 g/kg day 3
- Stop nephrotoxins/diuretics; repeat tap at 48 h; long-term quinolone prophylaxis

Hepatorenal syndrome-AKI

- No response to albumin, no shock/nephrotoxins; terlipressin + albumin; transplant

Key point: Albumin in SBP prevents renal failure and saves lives.

Ch 75 · Alcohol-related liver disease in ICU

Severe alcoholic hepatitis

- Jaundice < 8 weeks, AST > ALT; Maddrey ≥ 32 or MELD > 20

Treatment

- Prednisolone 40 mg × 28 days (STOPAH: short-term benefit); Lille ≥ 0.45 at day 7 → stop
- Nutrition 35–40 kcal/kg; thiamine; infection screening; NAC may help

Associated problems

- Alcohol withdrawal (lorazepam/oxazepam), Wernicke's, HE (lactulose, rifaximin)
- Abstinence support; early transplant discussion in selected patients

Key point: Exclude infection before starting steroids.

Ch 88 · Bowel obstruction

Causes

- SBO: adhesions, hernia, cancer, Crohn's; LBO: cancer, volvulus, stricture; Ogilvie's

Management

- Fluids, correct K, NG decompression; water-soluble contrast for adhesive SBO
- Frailty and goals of care

Surgery

- Strangulation signs, rising lactate, closed loop, failure after 48–72 h
- Caecum > 12 cm; stent as bridge for obstructing cancer; endoscopic decompression of sigmoid volvulus

Key point: Closed-loop obstruction is a surgical emergency.

Ch 94 · Biliary infections

Cholangitis (Tokyo 2018)

- Inflammation + cholestasis + imaging; grade III = organ dysfunction

Treatment

- Cultures, pip-taz (± vancomycin if healthcare-associated)
- Biliary drainage: ERCP within 24 h (ASAP in grade III); PTC if ERCP fails
- Antibiotics 4–7 days after source control

Acalculous cholecystitis

- Critically ill, TPN, burns; wall > 3.5 mm, fluid, sludge
- Antibiotics + percutaneous cholecystostomy

Key point: Severe cholangitis needs drainage, not just antibiotics.

Ch 96 · The open abdomen

Indications

- Damage control, abdominal compartment syndrome, re-look for peritonitis/ischaemia

ICU care

- NPWT with fascial traction; aim fascial closure within 7 days
- Replace protein-rich losses; early enteral nutrition; analgesia; monitor IAP

Complications

- Enteroatmospheric fistula, loss of domain, hernia, infection, hypothermia

Key point: Early fascial closure reduces fistulas and mortality.